Low-Power CMOS Operational Amplifier Design in 45nm Technology

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***Abstract*—Energy-efficient design is the backbone of modern electronics, crucial for portable devices and medical systems where every milliwatt counts. This paper presents the design and simulation of a low-power CMOS operational amplifier (OpAmp) implemented in a CMOS technology node, optimized for applications requiring ultra-low power consumption. Unlike conventional OpAmp designs that operate in strong inversion, this work leverages moderate inversion to achieve a trade-off between power efficiency and gain performance, making it highly suitable for sensor interfaces in wearable health monitoring devices, hearing aids, and telemetry systems. The proposed design features a cascade second stage for boosting the overall gain, a dynamically biased differential input stage for improved efficiency, and current mirrors to optimize tail current, ensuring stable operation at low power budgets. Simulated results in Cadence demonstrate an open-loop gain of $67\,\mathrm{dB}$, a gain-bandwidth product of $2.31\,\mathrm{MHz}$, and a power consumption of only $12.5\,\mu\mathrm{W}$, showcasing its feasibility for power-sensitive applications. By offering an implementation-based validation of OpAmp performance in the $45\,\mathrm{nm}$ node, this work closes the gap between high-tech low-power electronics' theoretical concepts and real-world applications.**

***Index Terms*—Cascade Amplifier, CMOS Operational Amplifier, Differential Input Stage, Dynamic Biasing, Gain-Bandwidth Product, Low-Power Design, Moderate Inversion, Tail Current Boosting.**

I. Introduction

A. Motivation

Driven by market demands for higher processor speeds, increased memory capacity, and lower power consumption, VLSI technology nodes have significantly shrunk over the years, thanks to continuous advancements in lithography [1]-[2]. While these scaling improvements have greatly benefited digital CMOS circuits, analog circuits face challenges due to reduced voltage headroom [3], which prevents transistors from operating in saturation. Additionally, transistors become more prone to leakage and short-channel effects [4].

B. Literature Review

Although advanced process technologies like the 45 nm node offer higher frequencies and lower power consumption [5], these benefits are not as effectively realized in analog designs as they are in digital circuits [6]. Analog circuits often require additional current sources to maintain performance, especially at lower supply voltages. Since the power consumption of a circuit is directly proportional to the supply voltage, VDD, and the total current Itotal, reducing power consumption requires minimizing either V_{DD}, I_{total} or both [7]:

$$P_{\mathrm{diss}} = V_{\mathrm{DD}} \cdot I_{\mathrm{total}}. \tag{1}$$

However, reducing the total current negatively impacts the dynamic range [8], and lowering V_{DD} makes achieving transistor saturation more challenging [9]. Additionally, scaling V_{DD} often conflicts with the threshold voltage, V_{TH}, of transistors [10]. These challenges demand innovative design techniques to balance power, performance, and efficiency. The key difficulty in designing low-power operational amplifiers is achieving high gain, bandwidth, and stability while keeping power consumption low [11]. Traditional op-amp designs often require trade-offs between power, speed, and noise performance, limiting their use in power-constrained systems. However, recent technological advancements have opened new possibilities, enabling smaller, more power-efficient devices without compromising performance [12].

Traditionally, MOSFETs operate in the strong inversion region, achieving maximum saturated current. However, to reduce power consumption, they can also function in the weak or moderate inversion regions [13] - [14]. The trade-off between weak and strong inversion is crucial for optimizing gain and cutoff frequency. Weak inversion significantly reduces power consumption but results in slower response times and larger area requirements. In contrast, strong inversion enables high-speed and compact designs but at the cost of increased power dissipation. Moderate inversion strikes the optimal balance, offering low power consumption while maintaining acceptable performance for energy-efficient applications [15].

In the subthreshold region, the drain current, I_{D}, can be expressed as [16]:

$$I_{\mathrm{D}} = \frac{W}{L}\mu_{\mathrm{e}}\left(q\frac{kT}{q}\right)\cdot\left[\sqrt{\frac{\epsilon_{\mathrm{Si}}}{2qN_{\mathrm{A}}\phi(0,V_{\mathrm{GS}})-\phi_{\mathrm{p}}}}\right]e^{\frac{q\phi(0,V_{\mathrm{GS}})}{kT}}\cdot\left(1-e^{-\frac{qV_{\mathrm{DS}}}{kT}}\right) \tag{2}$$

where I_{D} is the drain current, W/L is the width-to-length ratio of the MOSFET, μ is the mobility, e is the electron charge, kT/q is the thermal voltage, ϵ_{Si} is the permittivity of silicon, N_{A} is the acceptor concentration, $\phi(0, V_{\mathrm{GS}})$ is the

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potential at the gate-source interface, ϕ_{p} is the potential for p-type material, and V_{DS} is the drain-source voltage.

C. Novelty

This design is innovative because it uses a dynamically biased differential input stage that operates in the mild inversion zone, improving transconductance efficiency while preserving power limitations. Furthermore, to increase the gain by raising output resistance without appreciably increasing power dissipation, a cascaded second stage is used. Dynamic tail current adjustment is made possible by the incorporation of an improved current mirror architecture, which minimizes mismatch effects and guarantees stable operation at low voltages. These design decisions provide an effective and scalable solution for ultra-low-power applications by addressing the issues of lower voltage headroom, short channel effect, and low output current effects in a $45\,\mathrm{nm}$ CMOS technology. Where nobody has implemented the same design in $45\,\mathrm{nm}$ while working in the moderate inversion region.

II. Design Methodology

The design of low-power, low-voltage Op-Amps for ultra-low power consumption and high-performance applications necessitates careful selection of device parameters and topologies. The following sections delve into the key metrics and methodologies used to achieve the specifications.

A. Design Specifications

The target design parameters for the Op-Amp are outlined below, focusing on ensuring the device meets the low power and high performance requirements in Table I.

TABLE I
Specifications of the Operational Amplifier

Parameter	Specification
Supply Voltage (V_{DD})	$2.0\,\mathrm{V}$
Power Consumption (P_{diss})	$< 20\,\mu\mathrm{W}$
Open-Loop Gain (A_v)	$\geq 60\,\mathrm{dB}$
Unity-Gain Bandwidth (GBW)	$\geq 2.0\,\mathrm{MHz}$
Slew Rate (SR)	$\geq 2.5\,\mathrm{V}/\mu\mathrm{s}$
Phase Margin (PM)	$\geq 50^{\circ}$
Input Common Mode Range (ICMR)	Rail-to-Rail
Output Voltage Swing	Rail-to-Rail

B. Circuit Design Topologies

To meet the design specifications while keeping power consumption low, various circuit configurations are explored. The following topologies are used:

1) Push-Pull Operational Amplifier in Weak Inversion: Consider the push-pull operational amplifier configuration as shown in the Fig. 1. In this configuration, the differential pair consists of complementary MOS transistors (NMOS and PMOS), operating in weak inversion where the current is exponentially related to the gate-source voltage. The first stage provides moderate gain, while the second stage amplifies the output signal using current mirrors, ensuring high output drive capability with minimal power consumption.

The gain of the first stage is significantly low, offering little to no gain. However, the second stage contributes a substantial amount of gain. The total gain of the amplifier can be approximated by considering the transistors M3-M8, which act as current mirrors, and also the pairs M4-M6, and M7-M9. The total gain can be expressed as:

$$A_v = \frac{g_{m1} \cdot \left(\frac{S_6}{S_4}\right)}{(g_{ds6} + g_{ds7})} \tag{3}$$

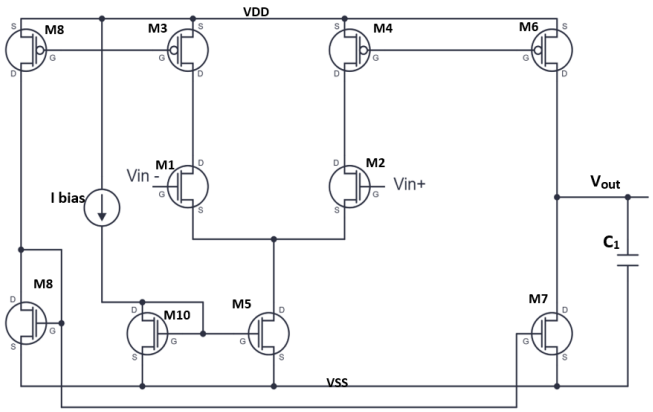


Fig. 1. Push-Pull Operational Amplifier Topology working in Moderate inversion region

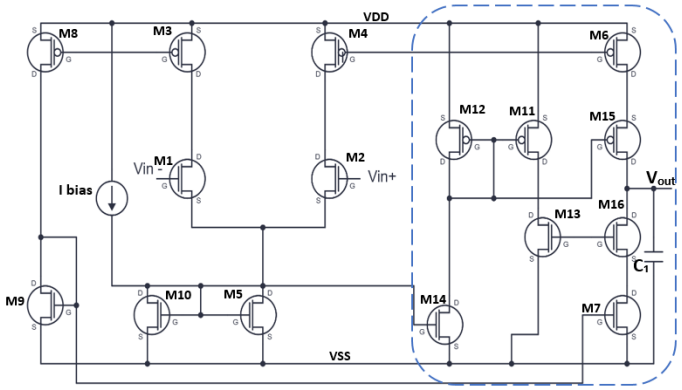


Fig. 2. Cascade Second Stage Topology for improved gain in weak inversion region

2) Cascade Second Stage for Gain Improvement: In the above design, Fig. 1, the gain was one of the issues in the subthresold region. To enhance the overall gain, a cascade second stage is implemented in the Fig. 2.

Cascoding in the second stage enhances the op-amp's gain by significantly increasing the output resistance, thereby it amplifying the overall voltage gain. It mitigates channel length modulation, preserving high intrinsic gain. Additionally, it improves Miller compensation, stabilizing frequency response.

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The gain of the overall opamp using the cascade stage can be described by the following expression:

$$A_v = \frac{I_{M5}}{2I_{M7}} \cdot \left(\frac{b}{n_n V_t^2}\right) \cdot \frac{1}{\left(\lambda_p^2 n_p + \lambda_n^2 n_n\right)} \tag{4}$$

Where b represents the ratio of the sizes of transistors M6 to M4. The parameters λ_n and λ_p are the channel-length modulation factors of the NMOS and PMOS transistors, respectively, while n_n and n_p are the subthreshold slope factors of the NMOS and PMOS transistors, respectively.

C. Design

Large output currents cannot be delivered by the amplifiers studied thus far while preserving low power consumption during quiet operation.

The literature has put forward an innovative answer to this problem [20]. Every time a differential input voltage is delivered, the method increases the tail current, which is eventually accessible at the output via current mirroring.

The proposed Op-Amp design integrates the concepts to optimize gain, power consumption, and maintain high output cuurents in the quiescent operation. The design of our circuit schematic is shown in Figure 3.

Let us consider a circuit that is the MOS M_{18} and M_{21} is similar to the MOS M_3 and M_4, and MOS $M_{27}, M_{22}, M_{25}, M_{26}, M_{24},\ M_{27}$ are all equal. Additionally, MOS M_{29} and M_{28} can be mathematically written in this manner:

$$\frac{W_{29}}{L_{29}} = A \cdot \left(\frac{W_{27}}{L_{27}}\right) \tag{5}$$

$$\frac{W_{28}}{L_{28}} = A \cdot \left(\frac{W_{26}}{L_{26}}\right) \tag{6}$$

The currents i_1 , i_2 are equal when the system is in a state of quiescence. M_{26} and M_{28} do not experience any current flow as a result of the current in M_{24} equaling the current supplied by M_{19}. No current also passes through M_{29}. No extra current is given to the differential stage in these circumstances.

When the circuit operates under non-quiescent conditions, if $v_{i1} > v_{i2}$, then $i_2 > i_1$, leads to an increase in the tail current that is proportionately provided to the input differential stage is $A(i_2 - i_1)$. On the other hand, $i_1 > i_2$ if $v_{i1} < v_{i2}$, which raises the tail current by $A(i_1 - i_2)$.

To derive the output current, assume the input differential voltage $V_{IN} = v_{i1} - v_{i2}$ is positive, implying that $i_2 > i_1$. The total current can be expressed as:

$$(1 + A)i_1 + (1 - A)i_2 = I_5 \tag{7}$$

By utilizing the weak inversion drain current relation and defining V_{IN}, the ratio between i_2 and $i_1$satisfying the following condition:

$$\ln\left(\frac{i_2}{i_1}\right) = \frac{V_{IN}}{nV_t} \tag{8}$$

where n is the subthreshold slope factor and V_t is the thermal voltage.

The output current, i_{OUT}, is defined as $b(i_2 - i_1)$, where b is a proportionality constant. Substituting the expressions for i_1 and i_2 yields:

$$i_{OUT} = \frac{bI_5\left[\exp\left(\frac{V_{IN}}{nV_t}\right) - 1\right] \cdot (1 + A)}{(A - 1) + \exp\left(\frac{V_{IN}}{nV_t}\right)} \tag{9}$$

In this circuit, M_{18} through M_{21} form part of the biasing network and are matched to M_3 and M_4 for symmetry. Transistors M_{22} to M_{27} act as current mirrors, ensuring proper scaling and current distribution, while M_{28} and M_{29} contribute to the additional tail current when the differential input voltage is non-zero. The exponential dependence of the output current on the differential input voltage results from the operation of the transistors in the weak inversion region.

III. Results and Discussion

This section presents the key simulation parameters of an OpAmp namely, AC analysis, slew rate, Power Supply Rejection Ratio (PSRR), and Common-Mode Rejection Ratio (CMRR). The evaluation of these parameters is essential to ensure the amplifier meets the desired performance targets in low-power applications.

A. AC Analysis

The AC analysis evaluates the frequency-dependent behavior of the operational amplifier, focusing on essential parameters such as gain, bandwidth, and phase margin. These metrics are crucial for ensuring stable and efficient operation, particularly in low-power applications where power efficiency often comes at the cost of reduced performance.

The performance metrics from the AC analysis are as follows:

- **Open-Loop Gain:** 66.85 dB
- **Gain Bandwidth Product (GBW):** 2.31 MHz
- **Phase Margin:** 57.78°

1) Analysis and Implications: Figure 4 illustrates the frequency response of the designed operational amplifier, which achieves an open-loop gain of 66.85 dB and a GBW of 2.31 MHz. This performance strikes a balance between gain and efficiency, making it suitable for low-frequency applications such as biomedical instrumentation and portable electronics, where speed is less critical. The phase margin of 57.78° ensures stable operation, providing resilience against transient disturbances and power supply fluctuations, essential for low-power designs.

B. Transient Response

The transient response of an operational amplifier is characterized by its ability to handle rapid changes in input signals as we can see in the Fig. 5. A key parameter here is the slew rate, which determines the maximum rate at which the output voltage can change. The slew rate is given by:

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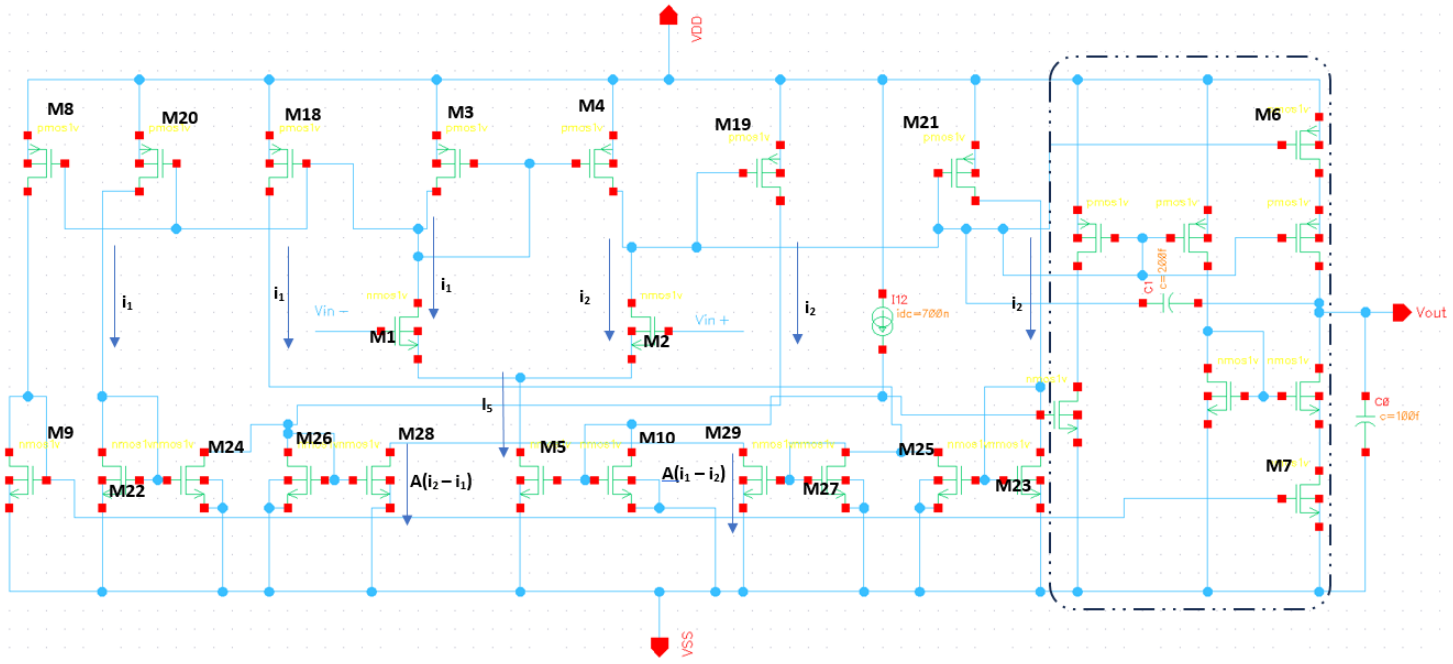


Fig. 3. Proposed Op-Amp Design Schematic operating in the subthresold region maintains high output current in the quiescent.

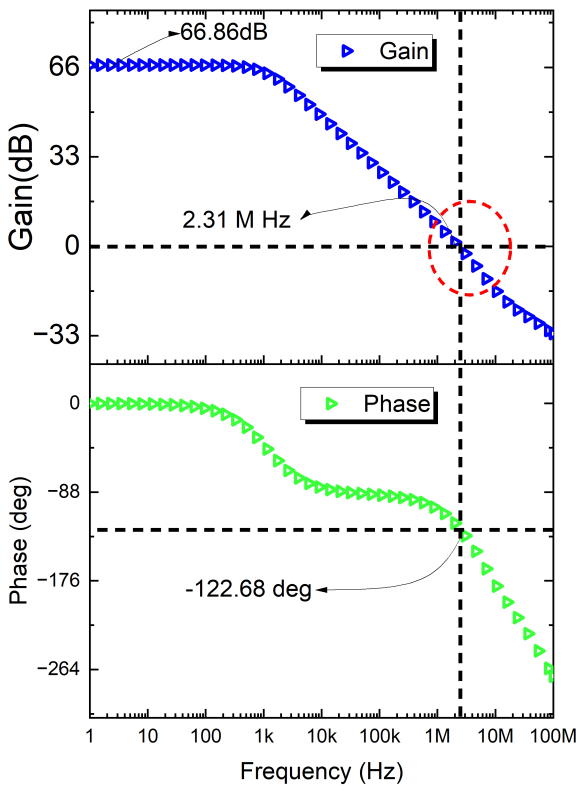


Fig. 4. Frequency response showing gain and phase characteristics

$$\text{Slew Rate} = \frac{I_{\text{tail}}}{C_L}$$

where I_{tail} is the tail current of the differential pair and C_L is the load capacitance at the output.

In this design, the measured slew rate is 2.5 V/µs, while the settling time is $0.89\,\mu s$.

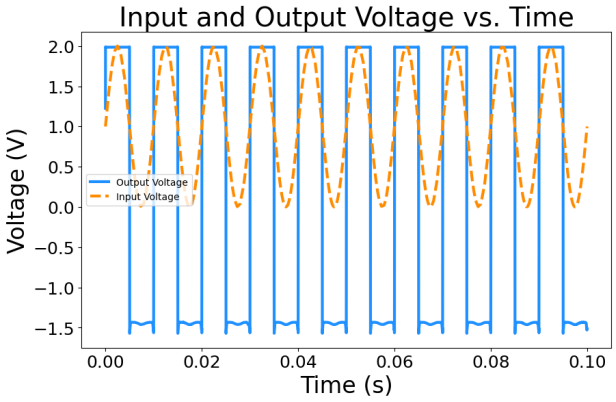


Fig. 5. Transient response

C. Power Supply Rejection Ratio (PSRR)

The Power Supply Rejection Ratio (PSRR) measures an operational amplifier's ability to reject fluctuations or noise in the supply voltage, ensuring stable output. It is defined as:

$$\text{PSRR} = 20 \cdot \log_{10}\left(\frac{\Delta V_{\text{PS}}}{\Delta V_{\text{OUT}}}\right)$$

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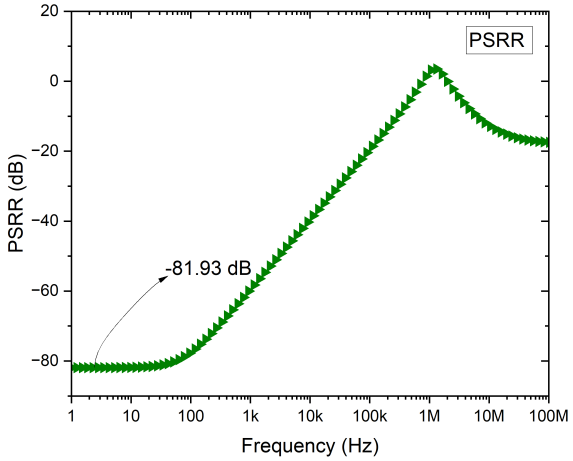


Fig. 6. PSRR analysis of the operational amplifier

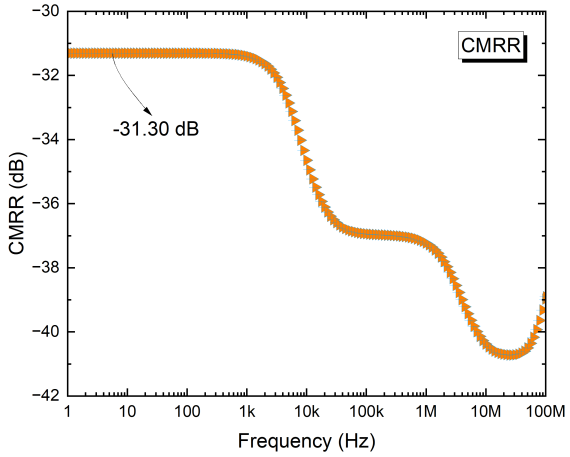


Fig. 7. CMRR analysis of the operational amplifier

where ΔV_{PS} is the change in power supply, and ΔV_{OUT} is the resulting voltage difference at output node. The simulated PSRR in Fig. 6 for this design is approximately 150 dB, indicating strong immunity to supply noise and enhanced signal integrity.

D. Common-Mode Rejection Ratio (CMRR)

The Common-Mode Rejection Ratio (CMRR) measures the amplifier's ability to reject common-mode signals affecting both inputs equally. It is defined as:

$$\text{CMRR} = 20 \cdot \log_{10}\left(\frac{A_{DM}}{A_{CM}}\right)$$

where A_{DM} and A_{CM} are the differential and common-mode gains, respectively. The simulated CMRR in Fig. 7 for this design is 100 dB, indicating good rejection of noise and interference, thereby improving signal fidelity.

TABLE II
COMPARISON OF PERFORMANCE METRICS WITH PREVIOUS WORKS

Metric	[21]	[22]	[23]	[24]	This Work
Gain (dB)	64	61.1	73.03	72.6	66.85
GBW (MHz)	2.15	1.094	28.4	18.97	2.31
Slew Rate (V/µs)	10	2.86	20.5	10	2.5
PSRR (dB)	120	108	99.7	87	150
CMRR (dB)	64.4	61	51.5	64	100
Power Consumption (mW)	4.30	0.017	1.23	2.13	0.0125

IV. CONCLUSION

This work presents the design of a low-power operational amplifier tailored for efficient performance in low-voltage environments using adaptive biasing techniques. The design process focused on key parameters, including open-loop gain, bandwidth, phase margin, and slew rate, ensuring the amplifier is optimized for power-sensitive applications.

The simulation results highlight the amplifier's strong performance, achieving an open-loop gain of 66.85 dB, a bandwidth of 2.31 MHz, and a gain-bandwidth product of 2.31 MHz. The phase margin of 68° further indicates its stability. With a slew rate of 2.5 V/µs, the amplifier can efficiently handle fast input variations, while the settling time of 0.89 µs ensures a quick response to signal changes.

Additionally, the amplifier demonstrates excellent rejection of power supply noise (PSRR of 150 dB) and common-mode interference (CMRR of 100 dB), which are essential for maintaining signal integrity in real-world, noisy environments.

In summary, the proposed low-power operational amplifier achieves an optimal balance between performance and energy efficiency. It is well-suited for applications in analog signal processing, especially in portable devices and energy-constrained systems where minimizing power consumption is critical.

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